

# Search for Optimal Pulse Charging Parameters for Li-ion Polymer Batteries Using Taguchi Orthogonal Arrays

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**Abstract**— Many methods have been proposed for faster and better battery charging with increased energy efficiencies. One of the most promising has been the pulse charge algorithm—also referred to as a pulse charging technique. Many researchers have investigated the impact of pulse charging on lithium-ion batteries; However, the combined impact of duty cycle and frequency of the pulse charge current on lithium-ion polymer batteries is seldom investigated. This work uses the Taguchi orthogonal arrays (OA) to search for optimal pulse charging parameters that will maximize battery charge and energy efficiencies while decreasing charge time. Confirmation experiments were performed and compared with the benchmark constant current-constant voltage (CC-CV) charging method. It was determined that by operating a pulse charger using optimal parameters, charge time was reduced by 47.6%, battery charge and energy efficiencies were increased by 1.5% and 11.3% respectively when compared with CC-CV charging method at a charge rate of 0.5C.

**Index Terms**— Battery ac impedance, charge efficiency, constant current – constant voltage (CC-CV), electrochemical impedance spectroscopy, fast charging, li-ion polymer, pulse charging, Taguchi

## I. INTRODUCTION

RECHARGEABLE batteries have become prevalent as energy storage devices for most electronic mobile devices compared to supercapacitors. This is due to their high energy density and low self-discharge rate. Primarily, lithium-ion (Li-ion) batteries have found their way into most of today's consumer electronics, such as tablets, notebooks, and smartphones, and they will continue, as predicted, to gain a greater market share [1], [2]. Their higher energy density and capacity, low self-discharge rate, and absence of memory effect make them ideal for consumer electronic applications.

Most smartphones, tablets, and most recently, notebooks, are using Li-ion polymer (LiPo) batteries. LiPo batteries use a high

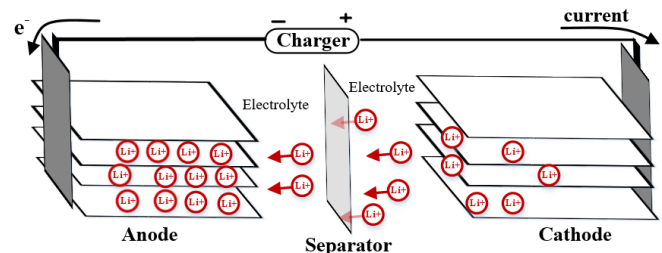


Fig. 1. Charge processes in a LiPo battery

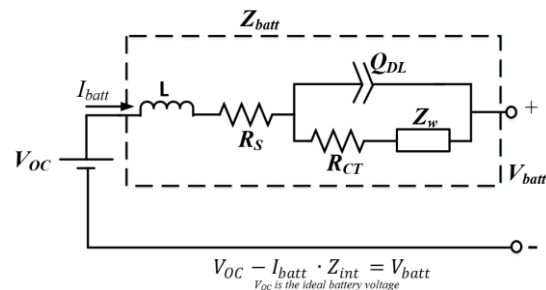


Fig. 2. LiPo battery model

conductivity polymeric semisolid to support the electrolyte and improve ion movement compared to Li-ion batteries which use liquid electrolytes [3]–[6]. LiPo batteries are lightweight, thinner, and much more flexible—able to fit into many form factors compared to their prismatic and cylindrical Li-ion counterparts [2]. LiPo batteries use metal oxide for their cathode and graphite for their anode. During charging, ions move from the cathode through the gel-based electrolyte saturated polymer separator to intercalate in the anode [7]–[9]. The reverse occurs during discharge. This is illustrated in Fig. 1.

During the charging of LiPo batteries, there are two well-known, particularly deleterious inherent effects that influence performance. These effects are dendrite formation and polarization. Dendrites are lithium deposits that can form on the surface of electrodes in the battery [3], [10]. If dendrites

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continue to grow, they can cause an internal short circuit that can lead to thermal runaway and potentially a fire or explosion [11]. Polarization can be described as the deviation of electrode potential across the battery from its stable value [6], [12], [13]. Concentration polarization, which is the buildup of a concentration gradient due to the intercalation of ions at the electrodes/electrolyte interface in the battery [12], [14], can lead to such a difference in potential. From an electrical perspective, polarization can be modeled [8] as the internal impedance,  $Z_{batt}$ , of the battery model shown in Fig. 2.  $R_{CT}$  is the charge transfer resistance representing the electrochemical charge transfer reaction,  $R_s$  is the electrolyte resistance,  $Z_w$  is the Warburg element denoting the mass transport effects in the battery,  $Q_{DL}$  is a constant phase element that models the double layer capacitance due to the charged area between the electrode and the electrolyte, and  $L$  is the inductance of the electrodes [15]–[17]. The impedance of  $Q_{DL}$  can be expressed as  $Z_{Q_{DL}} = 1/Q_{DL}(i\omega)^\alpha$ , where  $\omega$  is the angular frequency and  $\alpha$  is an empirical constant.

During charging, these polarization and dendrite formation effects must be considered due to the stringent charge requirements of the LiPo battery. As such, various charging techniques have been developed for lithium chemistry-based batteries, particularly Li-ion, in terms of improving battery health, life cycle, charge time, and charge efficiency [18]–[25]. Among these various techniques, the constant current – constant voltage (CC-CV) is the most widely adopted and is considered a benchmark when compared with other charging techniques [22].

The CC-CV charging technique has three charge phases: a) trickle charging phase, which uses a small charge current to precondition deeply discharged cells, b) a constant current phase which is initiated once the battery is out of the deeply discharged state and involves the use of a constant charge current  $I_{charge}$ , and c) a constant voltage phase where a constant voltage is employed with an associated decreasing charge current. The value of  $I_{charge}$  depends on the LiPo battery specifications and can range from 50% to as much as 320% of the rated charge current [26], [27]. The drawback of this charging technique is that the charging time is extended due to the constant voltage phase. The initial charging is much faster when a large current is used. However, to prevent overcharging and overvoltage, two features that can potentially damage the battery, the charge current must be reduced; therefore, charge time is extended.

Other works [19], [21], [23], [24], [28]–[30] have proposed various charging techniques which will enable fast charging and better battery charge efficiency. Commercial fast battery chargers [31] use charge currents greater than 100% of the rated current to enable fast charging until the battery voltage reaches a certain threshold. After this threshold, the charge current must be reduced to ensure safety and prevent overcharging and overvoltage. Depending on how fast the battery can be charged, much larger charger adapters might be needed. If a battery capacity is rated at 1000 mAh, it can realistically be charged in one hour, assuming no losses, by using a charge current of 1A.

Any shorter charge time will require a larger charge current, which implies thicker wires and higher wattage-rated adapters leading to an increase in cost.

Pulse charging, which entails pulsing a charge current in a controlled manner into the battery, is another technique to achieve fast charging and better battery charge efficiency [13], [24], [25], [32]–[34] without increased cost and complicated charge algorithms.

However, to get the benefits of pulse charging, the duty cycle and frequency must be properly selected. Different LiPo batteries have different impedance profiles; therefore, depending on the frequency at which the charge current is pulsed, the battery energy efficiency,  $\eta_{BE}$ , and battery charge efficiency,  $\eta_{BC}$ , will be different. Higher duty cycles should result in faster charging, but the probability of the battery being subjected to overcharging and overvoltage conditions is also high. It is, therefore, necessary to obtain the optimal charge parameters for pulse charging in order to maximize the  $\eta_{BE}$  and  $\eta_{BC}$  of the battery while reducing the charge time  $t_c$ . For a pulse charger, the battery charge parameters that can be controlled are charge current amplitude, frequency, and duty cycle. Previous pulse charger works have used charge frequencies equal to the frequency at which the impedance is at minimum [25] and 50% duty cycles [35]. Other works have also used varying duty cycles [13], [24] and frequencies [13] during pulse charging. [32] proposed charging at 75% duty cycle because any additional rest time of the charge pulses will contribute to increased charge time. It is apparent that there is a need to find the optimal charge parameters for pulse charging based on an optimization method in order to maximize charging efficiency and reduce charge time. It is also important to determine which factors and factor levels have the most impact on certain output responses and using this information can tune output responses to fit particular applications.

One way to determine these optimal parameters is to test all possible combinations, but this will increase manufacturing costs and consume time. It is, therefore, important to use optimization techniques to determine which charging parameters and levels have the greatest impact on  $\eta_{BE}$ ,  $\eta_{BC}$ , and  $t_c$ , and at the same time predict the performance of the pulse charger when dealing with manufacturing quality variability across batteries.

This paper proposes the use of a partial factorial design of experiments (DoE) method, i.e., the Taguchi orthogonal arrays (OA), to determine the optimal pulse charging parameters to maximize battery charge and energy efficiencies while reducing charge time. This technique can be applied to different battery capacities to obtain optimal charge parameters.

The paper is organized into five sections. Section II gives the background on pulse charging, whereas Section III gives an overview of Taguchi OA. Section IV describes the technique for searching for optimal pulse charging parameters. Experimental results are presented in Section V and conclusions are drawn in Section VI.

## II. BACKGROUND ON PULSE CHARGING

Pulse charging is an effective technique to decrease charge time and improve battery charge and energy efficiencies. Charging efficiency,  $\eta_{BC}$ , is defined as the ratio of the charge recovered from a battery under a defined load to the total charge consumed during its charging. Energy efficiency,  $\eta_{BE}$ , is defined as the ratio of the energy recovered from a battery to the energy needed for its recharge. Thus,  $\eta_{BC}$  and  $\eta_{BE}$  can be mathematically expressed in (1) and (2), respectively, as

$$\eta_{BC} = \frac{\int_0^{t_d} I_d dt}{\int_0^{t_c} I_c dt} * 100 \quad (1)$$

$$\eta_{BE} = \frac{\int_0^{t_d} V_d I_d dt}{\int_0^{t_c} V_c I_c dt} * 100 \quad (2)$$

where  $t_d$  is the discharge time,  $V_c$  and  $V_d$  are the average battery charge and discharge voltages [36], [37], and  $I_d$  and  $I_c$  are the discharge and charge current, respectively. During charging, if the rate at which the lithium ions ( $\text{Li}^+$ ) are electrochemically reduced at the graphite electrode is higher than the rate at which  $\text{Li}^+$  are intercalated into the graphite, then, a buildup will occur at the graphite-electrode interface, increasing concentration polarization [14]. Due to decreasing concentration and, hence, limited diffusion of  $\text{Li}^+$  during the later stage of charging, the constant voltage phase of a CC-CV charge technique uses decreasing charge current, thereby increasing charge time. However, this problem of prolonged charge time is avoided in pulse charging due to the relaxation periods in between pulses. This eliminates concentration polarization [38] and ensures equilibrium between the electrodes. In that regard, the next charge pulse is efficiently absorbed, thereby increasing power rate transfer, and hence decreasing charge time.

One of the characteristics of a pulse charge is its frequency. Depending on the frequency of the pulses, the response of the battery will be different. At extremely low frequencies, less than 1 Hz, the energy transfer in the battery is due to the charge transfer reaction. At medium frequencies, energy transfer is due to the surface charges on the electrodes. At high frequencies, above 1 kHz,  $Q_{DL}$  and  $L$  become significant [39].

The impedance of the battery is dependent on the frequency [13]. There is a point on the impedance magnitude plot at which a local minima occurs. The Warburg impedance,  $Z_w$ , is dependent on frequency. At high frequencies,  $Z_w$  is small due to the depth of diffusion reactants being shallow. Lower frequencies ( $< 1 \text{ Hz}$ ) cause diffusion reactants depth to increase, thereby increasing  $Z_w$ . If the pulse frequency is above 1 Hz,  $Z_w$  is negligible. In that case, the impedance,  $Z_{Batt}$ , of the battery model shown in Fig. 2, can be expressed as (3).

$$Z_{Batt} = R_s + \frac{R_{CT}(1 + \tau\omega^\alpha X)}{1 + 2\tau\omega^\alpha X + (\tau\omega^\alpha)^2} + j \left( \omega L - \frac{R_{CT}(\tau\omega^\alpha Y)}{1 + 2\tau\omega^\alpha X + (\tau\omega^\alpha)^2} \right) \quad (3)$$

where  $\tau = R_{CT}Q_{DL}$ ,  $X = \cos \frac{\alpha\pi}{2}$ , and  $Y = \sin \frac{\alpha\pi}{2}$ .  $Q_{DL}$  and  $R_{CT}$  form a low pass filter, which implies that above a certain frequency,  $f_{cut-off} = 1/(2\pi Q_{DL}R_{CT})$ , pulse charge current frequencies will be filtered, and this will vary for different

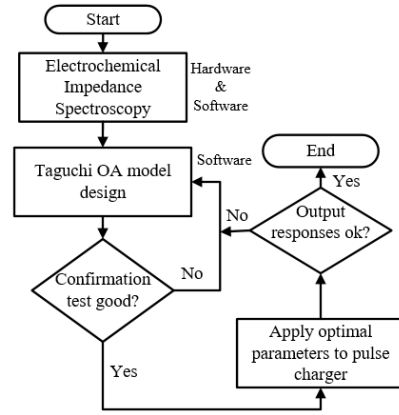


Fig. 3. Proposed method for search of pulse charging optimal parameters

batteries. It is important to perform pulse charging at a frequency  $f_{Zmin}$  where the battery impedance is minimum to reduce energy losses in the battery.

The duty cycle of the pulse charge is also of utmost importance. A higher duty cycle should supposedly speed up charging (as the duty cycle approaches 100% representing a constant current charge phase) but since it will be determined in this work, this is not valid in the case of pulse charging. Higher duty cycles could cause the battery to be prone to overcharging leading to the plating of the lithium metal, which can be hazardous. The discharge capacity of the battery can vary depending on the charge duty cycle [40]. One approach to improve charge efficiency and reduce charging time is to seek optimized parameters for the application of new charging pulses. This serves to minimize the parasitic consumption of energy associated with concentration polarization and inductive effects. This search for optimal pulse charging conditions was rendered manageable by employing a design of experimental approaches and the Taguchi OA method specifically.

## III. OVERVIEW OF TAGUCHI ORTHOGONAL ARRAYS

This section introduces the fundamentals of Taguchi OA methods, its benefits, and how they are designed.

### A. Taguchi OA overview

Experiment designs are different methods that are used to evaluate how different factors or parameters at different levels influence the output of a system simultaneously. By using statistical methods, predictive information about systems can be obtained.

Table I: Typical orthogonal array

| L | Factors                                       |                      |                | Y               |       |                 |
|---|-----------------------------------------------|----------------------|----------------|-----------------|-------|-----------------|
|   | F <sub>1</sub>                                | F <sub>2</sub> ----- | F <sub>n</sub> | Y <sub>a</sub>  | ----- | Y <sub>m</sub>  |
| 1 | A <sub>n</sub><br>Factors at different levels |                      |                | Y <sub>a1</sub> |       | Y <sub>m1</sub> |
| 2 |                                               |                      |                | Y <sub>a2</sub> |       | Y <sub>m2</sub> |
| ⋮ |                                               |                      |                | ⋮               |       | ⋮               |
| x |                                               |                      |                | Y <sub>ax</sub> |       | Y <sub>mx</sub> |

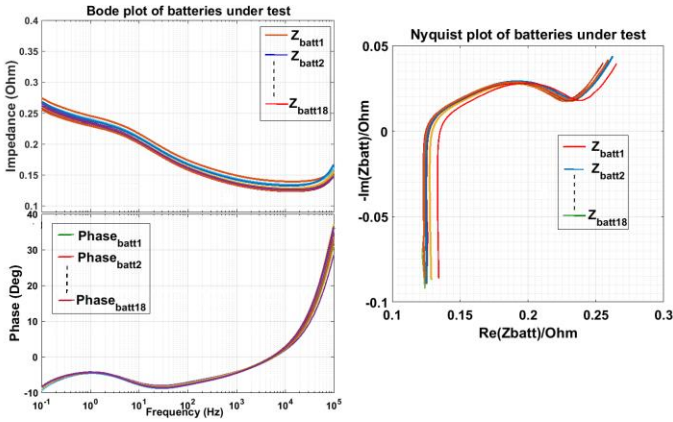


Fig. 4: Bode and Nyquist impedance plot of LiPo batteries under test

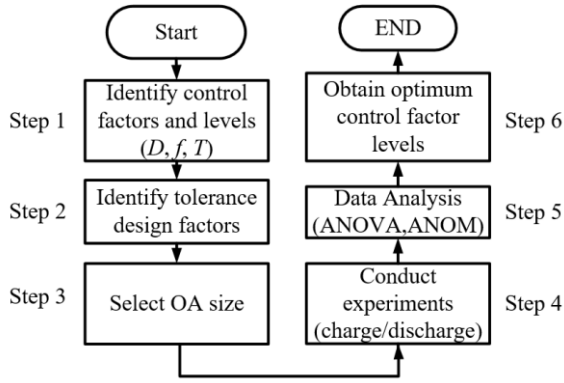


Fig. 5: Taguchi OA model design steps

Taguchi OA is one of the DoE methods for a robust experimental design named after Genichi Taguchi [41]. Taguchi OA focuses on achieving optimal settings for a series of control factors by the implementation of a limited number of experiments. It produces a balanced experiment where different levels of different factors are given an equal chance during the experiment's design and hence are weighted equally. This method is used to test how sensitive the output response of a system is to a set of control factors at different levels. Taguchi OA uses a balanced partial factorial design method to reduce the number of experiments needed. Even though the number of experiments are reduced, it has been proven that results from OA designs are comparable to that of full factorial designs [42], [43]. Taguchi OA has been successfully applied in the field of RF [44], electromagnetics [43], packaging [45], and power electronics [46]. [47] and [48] used Taguchi OAs to search for an optimal rapid charging pattern based on a multi stage constant current (MSCC) method for Li-ion batteries. The MSCC method is based on the premise that the charge current reduces progressively from one stage to another. [47] combined Taguchi OA with fuzzy logic to increase charge efficiency by 0.58% when compared to the CC-CV method. [48] used Taguchi OAs to find optimal current settings for each of the charging stages and thus increasing charging efficiency by an average of 0.83%.

### B. Design of Taguchi OA

The Taguchi OA design process is a three-step approach consisting of system design, parameter design, and tolerance design [49], [50]. System design refers to the design of a

Table II: Factors levels under test

| Factor         | Level |     |     |      |    |     |
|----------------|-------|-----|-----|------|----|-----|
|                | 1     | 2   | 3   | 4    | 5  | 6   |
| $D$            | 0.2   | 0.5 | 0.8 | -    | -  | -   |
| $T(^{\circ}C)$ | 0     | 23  | 45  | -    | -  | -   |
| $f$ (kHz)      | 0.1   | 1   | 6   | 12.6 | 50 | 100 |

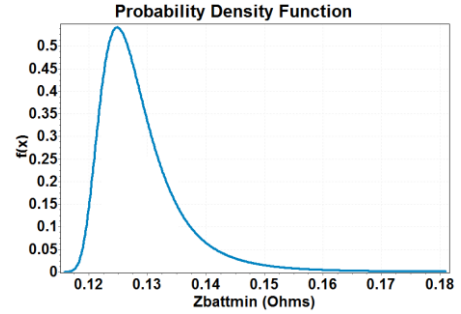


Fig. 6: PDF for  $|Z_{battmin}|$  of batteries under test

prototype by understanding what is needed; parameter design is recognizing the different variables or factors that will impact the design of the prototype, and tolerance design refers to taking into account unpredictable parameters which can impact the design of the system. In this paper, the system is the charger, the design parameters are i) the duty cycle of the pulses, ii) the frequency at which the pulse charger operates, and iii) the ambient temperature at which the li-ion battery is charged. Tolerance design could refer to statistical deviations in battery impedance parameters across different batteries.

Taguchi OA considers the factors that can be controlled, i.e. inner array factors, and that which cannot, i.e. outer array factors. A typical OA is shown in Table I where  $Y$  is the output response of the system. The OA can be expressed in the format  $L_x A^z$ , where  $L$  stands for Latin square,  $x$  is the number of rows in the array, corresponding to the number of experiments,  $A$  corresponds to the columns which represent the different levels of each factor, and  $z$  is the number of factors, i.e., control variables.

Signal-to-noise ( $S/N$ ) ratios, analysis of means (ANOM), and analysis of variance (ANOVA) can be used to analyze Taguchi OA experimental results. Taguchi  $S/N$  ratios can be expressed as

$$\frac{S}{N} = \begin{cases} -10 \log \left( n^{-1} \sum_i Y_{ij}^{-2} \right) & \text{larger - the - better} \\ -10 \log \left( n^{-1} \sum_i Y_{ij}^2 \right) & \text{smaller - the - better} \\ -10 \log \left( \frac{\bar{Y}^2}{v^2} \right) & \text{Nominal is the best} \end{cases} \quad (4)$$

where  $\bar{Y} = n^{-1} \sum_i Y_{ij}$  and  $v^2 = (n-1)^{-1} \sum_i (Y_{ij} - \bar{Y})^2$ ;  $i$  and  $j$  refer to the output response in the  $i_{th}$  row and  $j_{th}$  column.

### IV. SEARCH FOR OPTIMAL PULSE CHARGING PARAMETERS

This section describes the proposed system to determine optimal pulse charging parameters. Fig. 3 shows the flow chart of the proposed method for searching for the optimal parameters for pulse charging using Taguchi OA modeling and analysis.

Table III:  $L_{36}(3^2 \times 6^1)$  Orthogonal array

| Factors |   |   |   | Factors |   |   |   |
|---------|---|---|---|---------|---|---|---|
| No      | D | f | T | No      | D | f | T |
| 1       | 2 | 1 | 4 | 19      | 1 | 3 | 4 |
| 2       | 3 | 3 | 1 | 20      | 1 | 2 | 3 |
| 3       | 3 | 1 | 5 | 21      | 1 | 1 | 2 |
| 4       | 3 | 1 | 3 | 22      | 1 | 1 | 1 |
| 5       | 3 | 2 | 6 | 23      | 2 | 1 | 6 |
| 6       | 1 | 3 | 4 | 24      | 1 | 2 | 3 |
| 7       | 1 | 1 | 1 | 25      | 3 | 2 | 6 |
| 8       | 3 | 3 | 2 | 26      | 2 | 1 | 6 |
| 9       | 2 | 2 | 2 | 27      | 2 | 3 | 5 |
| 10      | 1 | 1 | 2 | 28      | 3 | 1 | 5 |
| 11      | 1 | 2 | 5 | 29      | 2 | 3 | 3 |
| 12      | 3 | 2 | 4 | 30      | 2 | 2 | 1 |
| 13      | 2 | 3 | 5 | 31      | 1 | 2 | 5 |
| 14      | 3 | 2 | 4 | 32      | 1 | 3 | 6 |
| 15      | 3 | 3 | 2 | 33      | 3 | 1 | 3 |
| 16      | 3 | 3 | 1 | 34      | 2 | 2 | 2 |
| 17      | 2 | 2 | 1 | 35      | 2 | 3 | 3 |
| 18      | 2 | 1 | 4 | 36      | 1 | 3 | 6 |

### A. Electrochemical Impedance Spectroscopy

The ac impedance spectrum of the batteries under testing must first be determined using electrochemical impedance spectroscopy (EIS). In this work, EIS was performed on 18 new 600 mAh 3.7 V batteries with similar electrochemical properties to determine the frequency at which battery impedance is minimized. The Bode and Nyquist impedance plot of LiPo batteries under test is shown in Fig. 4.

### B. Taguchi OA model design

Taguchi OA model design was then initiated to determine the optimal parameters for charging. Fig. 5 illustrates the following steps involved in the model design.

**Step 1.** Different pulse charging parameters or factors will result in different battery charger performances. The factors that can be controlled during pulse charging are duty cycle,  $D$ , frequency,  $f$ , of the charge current pulses, the amplitude of the charge current, and the ambient temperature at which the battery is charged,  $T$ . For a fair comparison to the CC-CV charging method, 50% of the rated charge current was used for all charging methods. The duty cycle levels were selected based on the pulse charger circuit, which had duty cycles ranging from 20% to 80%. Therefore, the maximum, minimum, and midpoint of the pulse charger duty cycle range was chosen, and hence the levels for the duty cycle were chosen to be 20%, 50%, and 80%. Selection of frequency levels was based on the change in  $Z_{batt}$  across the EIS spectrum. As frequency reduced, there was much more significant change in  $Z_{batt}$  compared to frequencies above the  $f_{zmin}$ . This is evident in Fig. 4. Six frequency levels, 0.1 kHz, 1 kHz, 6 kHz,  $f_{zmin}$ , 50 kHz, and 100 kHz, were chosen based on the change in impedance of 100 mΩ across the measurement frequency range as shown in Fig. 4. Taguchi OA is beneficial in this case as all factor levels do not need to be evaluated to determine which factor level has the most impact. The statistical methods from Taguchi OA analysis is comparable to full factorial designs [43]. The ambient temperature at which the battery is charged also has an impact on  $\eta_{BE}$ ,  $\eta_{BC}$ , and  $t_c$ . To evaluate the extent of this impact, the

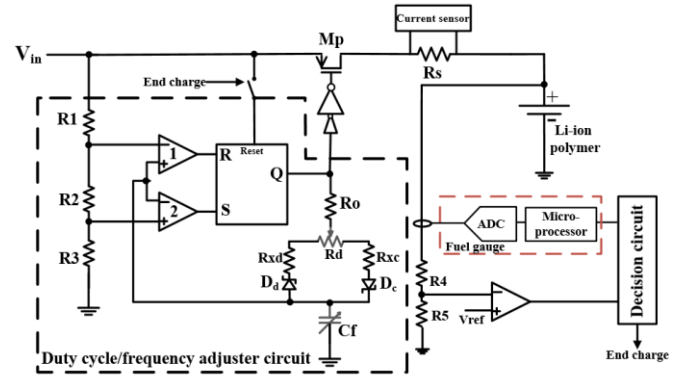


Fig. 7. Pulse charger schematic

temperature levels were set at 0 °C, 23 °C, and 45 °C. The factors and their levels are shown in Table II. Level 4 of  $f$  is  $f_{zmin}$ . The OA design is described in step 3.

**Step 2.** Tolerance design parameters were determined to be the impact of process parameters during manufacturing on battery impedance parameters. Fig. 6 shows the probability distribution function (PDF) for the minimum ac battery impedance  $Z_{battmin}$  of the batteries under testing. Since these manufacturing differences cannot be controlled in the experimental design, the search for optimal values must be able to account for and accommodate these variations.

**Step 3.** Based on three control factors, with two factors at three levels and the last factor at six levels, the mixed level OA was chosen to be  $L_{36}(3^2 \times 6^1)$  and shown in Table III. The OA was designed such that it was balanced, i.e., all the factor levels appear in the array for an equal number of times.

**Step 4.** The charging and discharging were both performed at charge/discharge current corresponding to 50% of the rated current in a controlled temperature chamber. The charging pulse parameters were applied at the specific temperatures per Table III. The battery charge and energy efficiencies were evaluated using (1) and (2) and charge time was recorded.

**Step 5.** Analysis of the evaluated results were used to determine the optimal factor levels to maximize battery charge and energy efficiencies while minimizing charge time. By using the objective function, the optimal factors that minimize battery charge time and maximize battery charge and energy efficiencies can be determined.

The effects of factor levels on the output responses,  $\eta_{BC}$ ,  $\eta_{BE}$ , and  $t_c$ , can be evaluated from the differences of the average  $S/N$  for the output responses corresponding to these levels and the overall mean. Effects of factor level,  $n$ , can be calculated using (5).

$$A_{Ln} = \left( \frac{1}{n_{Ln}} \sum_{i_{Ln}} Y_{i_{Ln}j} \right) - \bar{Y} \quad (5)$$

where  $A_{Ln}$  is the effect of factor level  $n$  on the output response,  $Y_j$ ,  $n_{Ln}$  is the number of output responses corresponding to level  $n$ , and  $i_{Ln}$  refers to the rows corresponding to level  $n$  responses only. The effects of the factors on the output responses can also be determined by the decomposition of their variance and computation of sum of squares. Sum of squares due to factor  $n$  can be expressed as

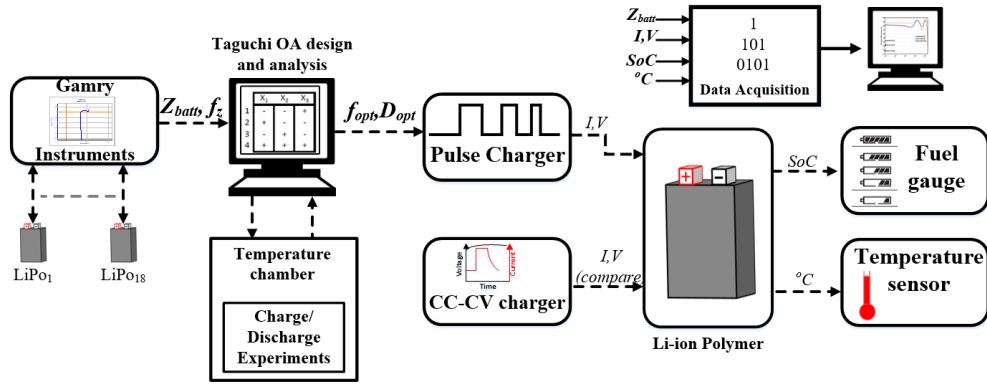


Fig. 8. Experimental test setup for searching for optimal pulse charger parameters for LiPo batteries

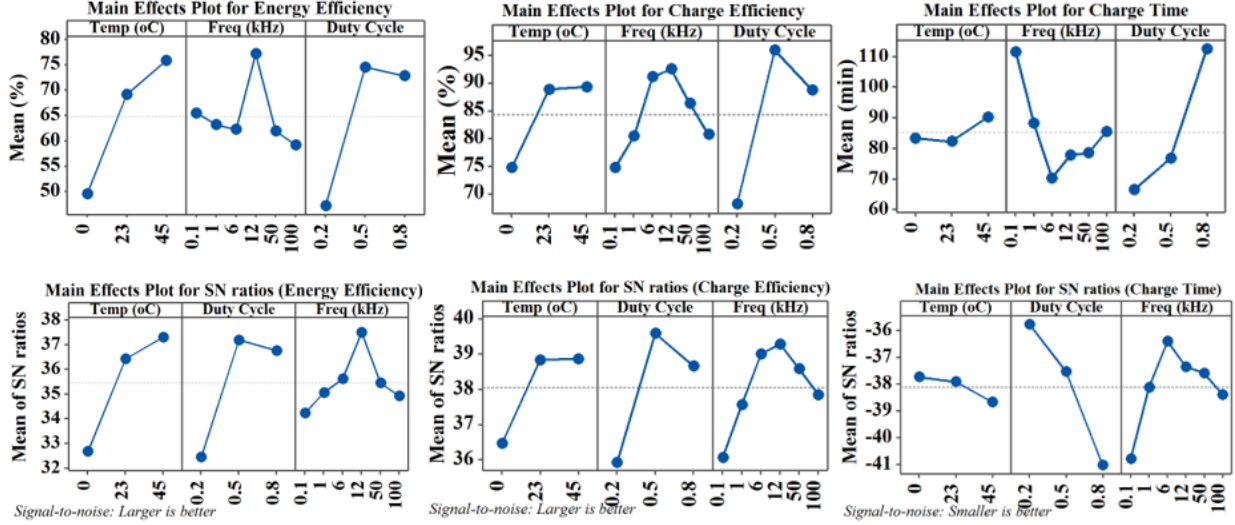


Fig. 9. Main effects plots for output responses

$$SS_{F_n} = \sum_{L_n} \left( n_{L_n} \cdot \left( \frac{1}{n_{L_n}} \sum_{i_{L_n}} Y_{i_{L_n}} - \bar{Y} \right)^2 \right) \quad (6)$$

Step 6. Optimal parameters were then obtained from these analyses and used to run confirmation tests for verification and validation.

### C. Pulse charger design

A pulse charger was designed such that the duty cycle could be changed without affecting frequency and vice versa. Fig. 7 shows the circuit diagram of the designed pulse charger. The pulse charger will take its voltage input from a current limited wall adapter,  $V_{in}$ , and pulse the charge current through the pass transistor,  $M_p$ , into the battery. The duty cycle and the frequency at which  $M_p$  is operated is controlled by the duty cycle-frequency adjuster circuit (DC-FAC). When a battery, serving as a device under test load, was placed in the circuit, the voltage detector and fuel gauge determine if the battery is fully charged or not. If it is fully charged, the pulse charger is switched off, otherwise, the DC-FAC is activated. The  $R$  input of the latch is low initially since  $V_{1+} < (V_{1-} = 2/3V_{in})$  for  $R1 = R2 = R3$ . The output,  $Q$ , goes high for  $V_{2+} > V_{2-}$ .  $C_f$  charges through  $D_c$  with a timing resistance of  $R_o + R_{d\_right} + R_{xc} + R_{DC}$ , where  $R_{d\_right}$  is the resistance to the right of the variable resistor and  $R_{DC}$  is the resistance of the diode. When  $V_{Cf}$  reaches  $2/3V_{in}$ , the  $R$  input of the latch goes high,  $Q$  is reset

to 0, starting the discharge phase of  $C_f$  through  $D_d$ ,  $R_{xd}$ ,  $R_{d\_left}$ , and  $R_o$ , where  $R_{d\_left}$  is the resistance to the left of the variable resistor. By varying the values of  $R_{d\_left}$  and  $R_{d\_right}$ , the duty cycle of the pulse charge current can be changed.

Since  $C_f$  is charged to only  $2/3V_{in}$ , the time constant during charging is  $T_{charge} = 0.7 (R_o + 1/(2R_d) + R_{xc})C_f$  and the time constant during discharging is  $T_{discharge} = 0.7 (R_o +$

Table IV: Output responses of Taguchi OA experiment

| Outputs |                 |                 |            | Outputs |                 |          |            |
|---------|-----------------|-----------------|------------|---------|-----------------|----------|------------|
| No      | $\eta_{BC}(\%)$ | $\eta_{BE}(\%)$ | $t_c(min)$ | No      | $\eta_{BC}(\%)$ | $\eta_I$ | $t_c(min)$ |
| 1       | 83.8            | 75.0            | 45.3       | 19      | 86.4            | 77.5     | 47.0       |
| 2       | 95.2            | 88.5            | 134.9      | 20      | 84.9            | 88.3     | 103.4      |
| 3       | 69.5            | 62.7            | 66.4       | 21      | 100.0           | 68.7     | 60.8       |
| 4       | 83.5            | 75.3            | 52.8       | 22      | 78.2            | 21.8     | 45.5       |
| 5       | 84.8            | 78.7            | 99.0       | 23      | 24.7            | 62.2     | 101.9      |
| 6       | 100.0           | 89.9            | 96.2       | 24      | 69.3            | 88.8     | 63.9       |
| 7       | 28.4            | 26.2            | 167.5      | 25      | 100.0           | 91.5     | 43.1       |
| 8       | 68.6            | 64.1            | 141.2      | 26      | 100.0           | 69.2     | 78.9       |
| 9       | 100.0           | 90.4            | 87.4       | 27      | 76.8            | 92.3     | 54.8       |
| 10      | 47.8            | 42.3            | 41.5       | 28      | 100.0           | 67.6     | 95.8       |
| 11      | 78.9            | 68.8            | 62.7       | 29      | 74.9            | 62.3     | 62.8       |
| 12      | 100.0           | 95.1            | 91.9       | 30      | 68.3            | 92.1     | 120.8      |
| 13      | 94.2            | 88.1            | 122.2      | 31      | 100.0           | 88.4     | 84.2       |
| 14      | 100.0           | 92.4            | 82.2       | 32      | 100.0           | 53.2     | 60.3       |
| 15      | 100.0           | 92.2            | 112.5      | 33      | 60.0            | 84.8     | 106.6      |
| 16      | 100.0           | 92.9            | 111.0      | 34      | 94.7            | 80.7     | 48.0       |
| 17      | 100.0           | 92.1            | 68.4       | 35      | 87.7            | 92.4     | 100.7      |
| 18      | 83.8            | 76.9            | 45.3       | 36      | 100.0           | 84.1     | 95.3       |

Table V. Charge improvement - Pulse charge vs CC-CV at 0.5C charge rate

| Output response | Charge method |       |                 |
|-----------------|---------------|-------|-----------------|
|                 | CC-CV         | Pulse | Improvement     |
| $\eta_{BE}$ (%) | 86.6          | 97.9  | 11.3% increase  |
| $\eta_{BC}$ (%) | 98.4          | 99.9  | 1.5% increase   |
| $t_c$ (min)     | 135.0         | 70.7  | 47.6% reduction |

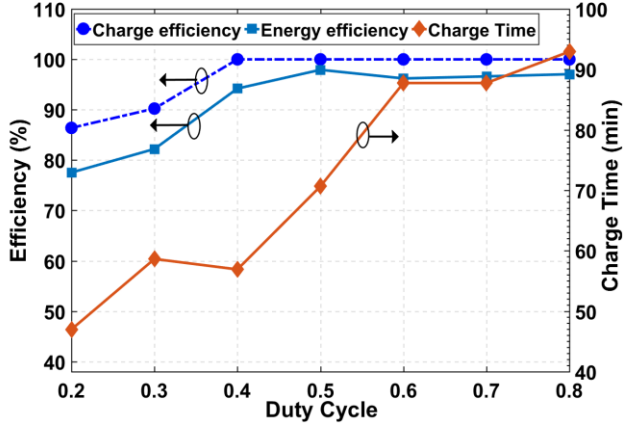


Fig. 10.  $\eta_{BC}$ ,  $\eta_{BE}$ ,  $t_c$  vs  $D$

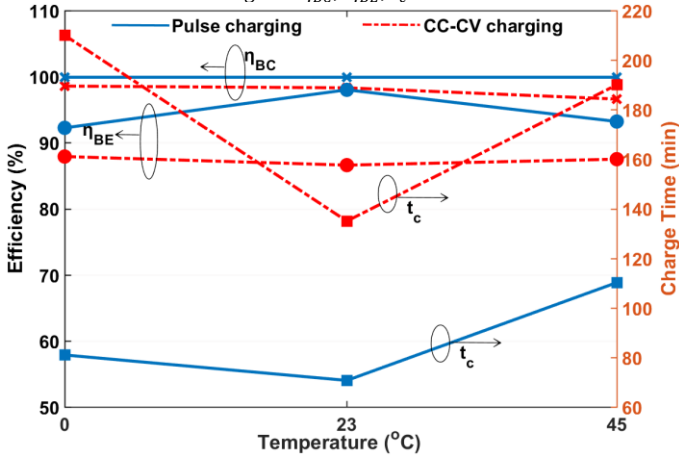


Fig. 11. CC-CV charge method vs Pulse charge method (50% duty cycle,  $f_{zmin}$ )

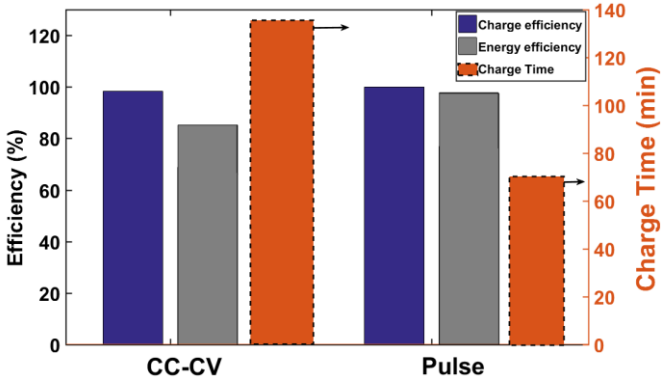


Fig. 12. CC-CV vs Pulse at room temperature (23°C) at 0.5C charge rate

$1/(2R_d) + R_{xd})C_f$ . The frequency,  $f$ , is, therefore, equal to  $1/1.4(R_o + 1/(2R_d) + R_{xd})C_f$ .

$Mp$  will be pulsed at the set frequency and duty cycle. LiPo batteries are sensitive to overcharge and any sort of overcharge can potentially damage the battery. To prevent overcharge and ensure safety, two methods of charge termination were

implemented. A fuel gauge was used to track the state of charge (SoC) of the battery and a voltage detector to track the battery voltage to prevent overvoltage. The fuel gauge terminates charging when SoC reaches 100%, while the voltage detector was set at 4.1V per battery data sheet specifications. The current sensor was used to monitor the charge current. The average charge current  $I_{avg}$  can be expressed as

$$I_{avg} = I_{pk} \cdot D \quad (7)$$

where  $I_{pk}$  is the peak pulse charge current amplitude.

## V. EXPERIMENTAL RESULTS

600 mAh 3.7 V LiPo batteries with a chemical composition of a graphite anode, lithium cobalt oxide cathode, lithium hexafluorophosphate electrolyte, and a microporous film blend of polypropylene and polyethylene for the separator [51] were used for all experiments, i.e. both CC-CV and pulse charging techniques.

The experimental setup is shown in Fig. 8. A Gamry Instruments Interface 1000E Potentiostat/Galvanostat/Zero Resistance Ammeter was used to determine the impedances of the batteries under test across a frequency range of 0.1 Hz to 100 kHz. This is shown in Fig. 4. The average  $Z_{battmin}$  was found to be 129 mΩ with an error margin of 0.18% across the minimum frequency range. The  $f_{zmin}$  of the batteries was determined to be at 12.6 kHz. The frequency information obtained from the EIS, together with the duty cycle of the pulse charger was used to select the levels of the factors to be used in the Taguchi OA design and analysis. The ambient temperature at which the battery was charged was also taken into consideration. Charge and discharge cycles were performed per Table III and recorded in Table IV.

Based on the results of the analysis, any output response can be maximized or minimized depending on the application. The objective of this work was to determine the factor levels that maximize  $\eta_{BC}$  and  $\eta_{BE}$  and minimize  $t_c$  of LiPo when undergoing pulse charging. By evaluating the output responses shown in Table IV using (6), ANOVA, the factor with the largest contribution to the total sum of squares can be determined. Duty cycle was found to make the largest contribution to all three output responses, 60.1% to  $\eta_{BE}$ , 61.8% to  $\eta_{BC}$ , and 68.0% to  $t_c$ . The frequency of operation of the pulse charger made the next largest contribution to  $t_c$ . Evaluating (4) and (5) using the output responses in Table IV, the optimal factor levels can be determined and plotted in Fig. 9. From this figure, the factor levels that correspond to the greatest impact on  $\eta_{BC}$  and  $\eta_{BE}$  are  $f = 12.6\text{kHz}$  ( $f_{zmin}$ ),  $D = 0.5$ , and  $T = 45^\circ\text{C}$ . Among these,  $f$  and  $D$  exhibit the greatest impact. Charging at  $f_{zmin}$  minimizes energy losses, thereby increasing  $\eta_{BC}$  and  $\eta_{BE}$ . In terms of the ambient temperature at which the battery was charged, as a general rule of thumb the rate of reaction doubles for every 10°C rise in temperature [52]. However, there was less than 3% change in  $\eta_{BC}$  and  $\eta_{BE}$  in going from 23°C to 45°C. The factor levels that had the greatest impact on  $t_c$  were  $f = 6\text{kHz}$ ,  $D = 0.2$ , and  $T = 23^\circ\text{C}$ . With efficiency and charge time having different optimal factor level values, the user can determine which output responses are paramount and select that which is suitable for the application.

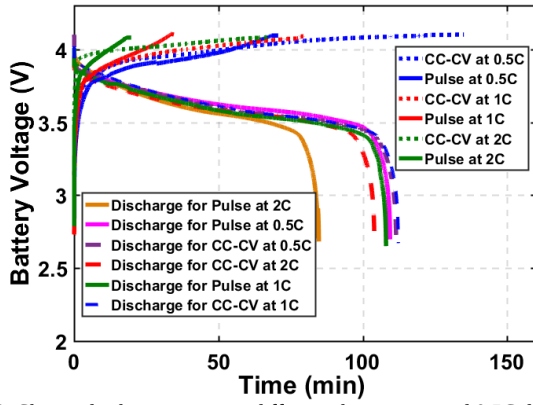


Fig. 13. Charge-discharge curves at different charge rates and 0.5C discharge rate

In this work, high efficiency is important as the discharge of the battery, such as usage of a cell phone, which can be prolonged after charging. Operating the pulse charge at 20% duty cycle and 6 kHz results in very poor  $\eta_{BE}$ . The difference in charge time is +10 minutes and +15% for  $\eta_{BE}$  when comparing charging at  $f = 6\text{ kHz}$ ,  $D = 0.2$ ,  $T = 23^\circ\text{C}$  to charging at  $f = 12.6\text{ kHz}$  ( $f_{zmin}$ ),  $D = 0.5$ ,  $T = 45^\circ\text{C}$ . By using S/N ratios (4), the larger-the-better for  $\eta_{BC}$  and  $\eta_{BE}$  and the smaller-the-better for  $t_c$ , the factor level effects can also be seen in Fig. 9. Regarding the ambient temperature at which the battery is charged, charging at  $0^\circ\text{C}$  produced poor  $\eta_{BC}$  and  $\eta_{BE}$  and longer charge time. This is due to the reduced mobility of the  $\text{Li}^+$ .

A confirmation experiment was run at  $f_{zmin}$  across various duty cycles and is presented in Fig. 10. This confirms what was previously observed in Fig. 9, i.e., charge time increases with  $D$ , and maximum  $\eta_{BE}$  occurs at  $f_{zmin}$ . The increase in  $t_c$  as the duty cycle increased was due to the LiPo battery not having enough rest between pulses for effective ion redistribution, thereby contributing to concentration polarization and preventing the maximum charge from being absorbed.

Using the optimal factor levels of  $f = 12.6\text{ kHz}$  ( $f_{zmin}$ ) and  $D = 0.5$  to maximize  $\eta_{BC}$  and  $\eta_{BE}$ , the pulse charge method was compared with the CC-CV charge method across temperature at a charge and discharge rate of 0.5C and is shown in Fig. 11. The peak pulse charge current amplitude was 600 mA, per equation (7) and the charge current in the CC phase of the CC-CV charge method was 300 mA. The batteries were allowed to rest for about an hour and then discharged at a constant current of 300 mA, i.e. rate of 0.5C. Across all output responses, pulse charging produced better results. These improvements are summarized in Table V for a charge rate of 0.5C. Fig. 11 and Fig. 12 confirms what is indicated in Fig. 9 concerning charge time, i.e. charging at room temperature produced the minimum charge time. The longer charge time of the CC-CV charge method is due to the CV stage where a decreasing charge current was used to prevent overcharging. It is observed that  $\eta_{BE}$  for both charging methods in Table V are lower than  $\eta_{BC}$ . This is attributed to the voltage efficiency of the battery, the ratio of the average battery discharge voltage to the average battery charge voltage [36], [37], which is affected by polarization and ohmic losses. From Fig. 2,  $V_{batt}$  will be higher during charging and lower during discharging. Using the

Table VI. Output responses at the different charge rates

| Output response | 1C    |       | 2C    |       |
|-----------------|-------|-------|-------|-------|
|                 | CC-CV | Pulse | CC-CV | Pulse |
| $\eta_{BE}$ (%) | 83.0  | 90.6  | 85.4  | 79.8  |
| $\eta_{BC}$ (%) | 91.1  | 99.2  | 88.9  | 96.1  |
| $t_c$ (min)     | 79.7  | 33.5  | 66.8  | 19.8  |

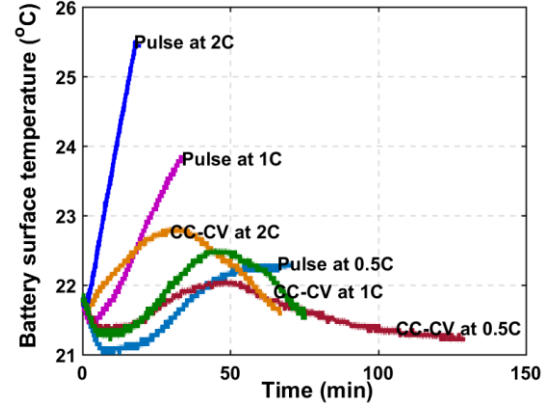


Fig. 14. Battery surface temperature during charging at different charge rates  
proposed pulse charging method reduced these losses contributing to a much higher  $\eta_{BE}$  when compared to the benchmark method.

Compared with [47] and [48], this work produced a better charge efficiency improvement when compared with the CC-CV charge method, a 1.5% increase compared to a 0.58% increase in [47] and 0.83% increase in [48].

The proposed pulse charging method operating at selected optimal parameters was also compared with the CC-CV charge method at the charge rates of 1C and 2C. The peak pulse charge current amplitudes were 1200 mA and 2400 mA for charge rates of 1C and 2C respectively, per equation (7). In the CC-CV charging method, the CC phase charge current was 600 mA and 1200 mA with respect to the charge rates of 1C and 2C. Table VI summarizes the results. Fig. 13 shows the charge-discharge curves, and Fig. 14 shows the battery surface temperature characteristics during charging for the two charging methods at different charging rates. It is noticed that across all the different charge rates, the proposed pulse charging method produced a much faster charge time but at a slightly higher battery surface temperature when compared to the CC-CV method.

At a charge rate of 2C, the proposed pulse charger produced a charge time reduction of 70.4% but at a rapidly increasing battery surface temperature as shown in Fig. 14 and a reduced discharge capacity as shown in Fig. 13. The 18.4% reduction in discharge capacity, when compared with that of the battery subjected to a CC-CV charging at 2C, was due to the early termination of charging to prevent overvoltage conditions due to the 600 mAh LiPo battery being subjected to a high peak current of 2400 mA. At this high charging rate, the life cycle of the battery can be negatively impacted [53].

The initial drop in battery surface temperature during charging was due to the endothermic reactions [54] which become overwhelmed with other sources of heat as charging

continues and hence, the increase in temperature. The higher temperature rise is more noticeable at the charge rates of 1C and 2C. Although this temperature rise is within the Japan Electronics and Information Technology Industries Association (JEITA) guidelines for battery charging safety in the standard temperature range [55], the increased charge rate can lead to reduced battery life cycle [53].

By using the pulse charger at the optimal settings,  $\eta_{BC}$  and  $\eta_{BE}$  improved by 1.5% and 11.3% respectively, and charge time was reduced by 47.6% at a charge rate of 0.5C. Additionally, at a charge rate of 1C,  $\eta_{BC}$  and  $\eta_{BE}$  improved by 8.1% and 7.6% respectively, and charge time was reduced by 58.0%.

## VI. CONCLUSION

This paper proposes a new technique to experimentally search for the optimal parameters for the pulse charging of a LiPo battery by using Taguchi orthogonal arrays. This approach can be extended to both large battery capacities, such as those used in electric vehicles, and smaller battery capacities, such as those used in portable electronic applications. This technique provides the optimal parameters that maximize battery charge and energy efficiency while decreasing charge time. The approach can also be extended to select factors and factor levels that maximize or minimize any preferred output response. The design of the Taguchi OA, the experimental procedure, the design of the pulse charger, and analysis of the output responses data have been presented. The optimal parameters for the pulse charging of a 3.7V 600 mAh LiPo battery was determined to be 50% duty cycle, which gives an equal time for the battery to absorb and redistribute charge efficiently, a frequency equal to the frequency at which the minimum ac impedance of the battery occurs, which minimizes the energy losses in the battery, and room temperature, which is ideal for most applications. When compared with the benchmark CC-CV charge method, operating the pulse charger at these parameters increased battery energy and charge efficiencies by 11.3% and 1.5% respectively, and decreased charge time by 47.6%. This work has demonstrated that by using the methods presented herein, optimal parameter values of a pulse charger can be determined.

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